Melissa Fischer

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SUMMARY

Software engineering student at 42 Porto with a strong foundation in **C/Linux**, currently building skills in **Python** and **C++**. Over the past year, I have been developing projects at 42 Porto with a focus on code quality, clean structure, memory management and rigorous testing to ensure reliable program behavior. Through collaborative projects, I have strengthened my use of Git, version control workflows, Agile practices, and peer feedback through code reviews. Interested in **Software Engineer** roles where strong problem-solving, clean code, collaboration and continuous learning are essential.

PROJECTS

Cub3D • 42 Porto • 2026

- Built a 3D raycasting engine in C inspired by Wolfenstein 3D, rendering a dynamic first-person view inside a maze.
- Implemented map parsing, player movement, wall collision handling, and real-time navigation logic for a playable maze experience.
- Focused on debugging/testing, clean code and leak-free execution to ensure the project met 42 quality standards.

Philosophers • 42 Porto • 2026

- Simulated the Dining Philosophers problem in C using POSIX threads (pthreads) and mutexes, with zero data races and leak-free execution verified with Valgrind tool.
- Managed concurrent resource between N philosophers with mutex locking to prevent deadlocks and created monitor thread to get accurate simulation stop.
- Applied a staggered start strategy (odd philosophers think first) to prevent deadlock and starvation from the start, with threads joined and mutexes destroyed on clean exit.

Minishell • 42 Porto • 2026

- Engineered a UNIX shell in C with command parsing, tokenization, redirections, signal handling and 5 built-ins.
- Managed code progress as a team with Git and Agile workflow: 20+ pull requests and 15+ conflict resolutions in a collaborative team setting.
- Deepened understanding of how a shell work behind the scenes through process forking, execve execution and managing environment variables with linked list structures.

EXPERIENCE

Animation Workshop Leader • TUMO • September 2025 - Present • Porto, Portugal

- Directed hands-on animation workshops from concept to final project using Toon Boom Harmony.
- Delivered curriculum to 120+ students/semester with 85% project completion rate; oversaw ~15 students/workshop adapting to different learning speeds.

ACTIVITIES

- **Game Dev Club Coordination • 42 Porto • Present:** Organizing events and projects. Currently working on an educational game project for kids about self-protection on the internet in partnership with Cybersecurity Club.
- **2nd Place, Hackathon • Blip • 2025:** Coding challenges in front-end, back-end, AI and User Experience (UX).
- **2nd Place, GameJam • 42 Porto • 2025:** Made 3D art and imported assets in Godot for a first person game. Worked under pressure with tight deadline and version control.
- **1st Place, 3D Modelling and Augmented Reality • WorldSkills Portugal • 2024:** Project developed in Blender, Substance Painter and Unity. Working with complex textures and particle systems to deliver a high-quality 3D asset.

EDUCATION

Software Development Programme • 42 Porto • 2025 - Present

- Peer-to-peer programming school with project-based learning. Focused in C, C++, Bash, Unix, algorithms, data structures and memory management.

Python Discovery Piscine Bootcamp • 42 Porto • 2026

Bachelor's Degree in Biology • University of Aveiro • 2022

LANGUAGES

Portuguese (Native) - English (C1) - French (Basic - Currently studying)